

Object Explorer | Dashboard | Properties | SQL | Statistics | Dependencies | Dependents | Processes | airport_DB/emilakshabayev@s*

airpot_DB/emilakshabayev@s

Query | Query History

```
1 --1
2 create index if not exists idx_flights_act_departure_time
3 on flights (actual_departure);
4
5 --2
6 create unique index if not exists uq_flights_fno_sched
7 on flights (flight_no, sch_departure_time);
8
9 --3
10 create index if not exists idx_flights_dep_arr_airports
11 on flights (departing_airport_id, arriving_airport_id);
12
13 --4
14 explain analyze
15 select f.flight_id
```

Data Output | Messages | Notifications

CREATE INDEX

Query returned successfully in 38 msec.

Total rows: Query complete 00:00:00.038 LF Ln 4, Col 1

Object Explorer: Columns (14): flight_id, flight_no, scheduled_departure, scheduled_arrival, departure_airport_id, arrival_airport_id, departing_gate, arriving_gate, airline_id, status, actual_departure, actual_arrival, created_at, update_at

Object Explorer | Dashboard | Properties | SQL | Statistics | Dependencies | Dependents | Processes | airport_DB/emilakshabayev@s*

airpot_DB/emilakshabayev@s

Query | Query History

```
1 --1
2 create index if not exists idx_flights_act_departure_time
3 on flights (actual_departure);
4
5 --2
6 create unique index if not exists uq_flights_fno_sch
7 on flights (flight_id, scheduled_departure);
8
9
10 select flight_no, scheduled_departure, count(*) as duplicates
11 from flights
12 group by flight_no, scheduled_departure
13 having count(*) > 1;
14 delete from flights f
15 where ctid not in (
```

Data Output | Messages | Notifications

CREATE INDEX

Query returned successfully in 80 msec.

Total rows: Query complete 00:00:00.080 LF Ln 9, Col 1

Object Explorer: Columns (14): flight_id, flight_no, scheduled_departure, scheduled_arrival, departure_airport_id, arrival_airport_id, departing_gate, arriving_gate, airline_id, status, actual_departure, actual_arrival, created_at, update_at

Object Explorer: Constraints, Indexes, RLS Policies, Rules, Triggers, passengers

Object Explorer | Dashboard | Properties | SQL | Statistics | Dependencies | Dependents | Processes | airport_DB/emilakshabayev@s*

airpot_DB/emilakshabayev@s

Query | Query History

```

6 create unique index if not exists uq_flights_fno_sch
7 on flights (flight_id, scheduled_departure);
8
9
10 --3
11 create index if not exists idx_flights_dep_arr_airports
12 on flights (departure_airport_id, arrival_airport_id);
13
14 --4
15 explain analyze
16 select f.flight_id
17 from flights f
18 where f.departing_airport_id = 101
19 and f.arriving_airport_id = 202;

```

Data Output | Messages | Notifications

CREATE INDEX

Query returned successfully in 30 msec.

Total rows: Query complete 00:00:00.030

✓ Query returned successfully in 30 msec. X

LF Ln 13, Col 1

Object Explorer | Dashboard | Properties | SQL | Statistics | Dependencies | Dependents | Processes | airport_DB/emilakshabayev@s*

airpot_DB/emilakshabayev@s

Query | Query History

```

10 --3
11 create index if not exists idx_flights_dep_arr_airports
12 on flights (departure_airport_id, arrival_airport_id);
13
14 --4
15 explain analyze
16 select f.flight_id
17 from flights f
18 where f.departure_airport_id = 101
19 and f.arrival_airport_id = 202;
20
21 --5
22 explain analyze
23 select f.flight_id, f.sch_departure_time, f.act_arrival_time
24 from flights f

```

Data Output | Messages | Notifications

Showing rows: 1 to 4 | Page No: 1 of 1

QUERY PLAN

1	Index Scan using idx_flights_dep_arr_airports on flights f (cost=0.28..8.29 rows=1 width=4) (actual time=0.239..0.240 rows=0 loops=...
2	Index Cond: ((departure_airport_id = 101) AND (arrival_airport_id = 202))
3	Planning Time: 4.319 ms
4	Execution Time: 0.755 ms

Total rows: 4 Query complete 00:00:00.044

LF Ln 20, Col 1

airport_DB/emilakshabayev@s

Query

```

27 from flights f
28 where f.departure_airport_id = 101
29 and f.arrival_airport_id = 202;
30 --5
31 explain analyze
32 select f.flight_id, f.scheduled_departure, f.actual_arrival
33 from flights f
34 where f.departure_airport_id = 101
35 and f.arrival_airport_id = 202
36 order by f.scheduled_departure desc
37 limit 50;
38 --6
39 select passport_number, count(*) as cnt
40 from passengers
41 group by passport_number
42 having count(*) > 1;
43 create unique index if not exists uq_passengers_passport
44 on passengers (passport_number);
45 select indexname, indexdef
46 from pg_indexes

```

Query History

QUERY PLAN

Step	Plan
1	Limit (cost=8.30..8.30 rows=1 width=12) (actual time=0.039..0.039 rows=0 loops=1)
2	Sort (cost=8.30..8.30 rows=1 width=12) (actual time=0.038..0.038 rows=0 loops=1)
3	Sort Key: scheduled_departure DESC
4	Sort Method: quicksort Memory: 25kB
5	Index Scan using idx_flights_dep_arr_airports on flights f (cost=0.28..8.29 rows=1 width=12) (actual time=0.019..0.019 rows=0 loops=1)
6	Index Cond: ((departure_airport_id = 101) AND (arrival_airport_id = 202))
7	Planning Time: 2.054 ms
8	Execution Time: 0.295 ms

Data Output

Showing rows: 1 to 8

Page No: 1 of 1

Total rows: 8 Query complete 00:00:00.034

airport_DB/emilakshabayev@s

Query

```

26 and f.arrival_airport_id = 202
27 order by f.scheduled_departure desc
28 limit 50;
29 --6
30 select passport_number, count(*) as cnt
31 from passengers
32 group by passport_number
33 having count(*) > 1;
34 create unique index if not exists uq_passengers_passport
35 on passengers (passport_number);
36 select indexname, indexdef
37 from pg_indexes

```

Query History

QUERY PLAN

Step	Plan
1	Limit (cost=8.30..8.30 rows=1 width=12) (actual time=0.039..0.039 rows=0 loops=1)
2	Sort (cost=8.30..8.30 rows=1 width=12) (actual time=0.038..0.038 rows=0 loops=1)
3	Sort Key: scheduled_departure DESC
4	Sort Method: quicksort Memory: 25kB
5	Index Scan using idx_flights_dep_arr_airports on flights f (cost=0.28..8.29 rows=1 width=12) (actual time=0.019..0.019 rows=0 loops=1)
6	Index Cond: ((departure_airport_id = 101) AND (arrival_airport_id = 202))
7	Planning Time: 2.054 ms
8	Execution Time: 0.295 ms

Data Output

Showing rows: 1 to 8

Page No: 1 of 1

Total rows: 0 Query complete 00:00:00.063

Object Explorer | Dashboard | Properties | SQL | Statistics | Dependencies | Dependents | Processes | airport_DB/emilakshabayev@s*

airport_DB/emilakshabayev@s

Query | Query History

```

32 from passengers
33 group by passport_number
34 having count(*) > 1;
35
36 create unique index if not exists uq_passengers_passport
37 on passengers (passport_number);
38
39 select indexname, indexdef
40 from pg_indexes
41 where schemaname = 'public' and tablename = 'passengers';
42
43 insert into passengers (passenger_id, first_name, last_name, date_of_birth, gender,
44 country_of_citizenship, country_of_residence, passport_number,
45 created_at, updated_at)
46 values (90001, 'test', 'ok', date '1990-01-01', 'Male',

```

Data Output | Messages | Notifications

CREATE INDEX

Query returned successfully in 40 msec.

Total rows: Query complete 00:00:00.040

✓ Query returned successfully in 40 msec. X

LF Ln 38, Col 1

Object Explorer | Dashboard | Properties | SQL | Statistics | Dependencies | Dependents | Processes | airport_DB/emilakshabayev@s*

airport_DB/emilakshabayev@s

Query | Query History

```

35
36 create unique index if not exists uq_passengers_passport
37 on passengers (passport_number);
38
39 select indexname, indexdef
40 from pg_indexes
41 where schemaname = 'public' and tablename = 'passengers';
42
43 insert into passengers (passenger_id, first_name, last_name, date_of_birth, gender,
44 country_of_citizenship, country_of_residence, passport_number,
45 created_at, updated_at)
46 values (90001, 'test', 'ok', date '1990-01-01', 'Male',
47 'Kazakhstan', 'Kazakhstan', 'KZ-PASS-00090001', now(), now());
48
49 insert into passengers (passenger_id, first_name, last_name, date_of_birth, gender,

```

Data Output | Messages | Notifications

Showing rows: 1 to 4 | Page No: 1 of 1

	indexname name	indexdef text
1	passengers_pkey	CREATE UNIQUE INDEX passengers_pkey ON public.passengers USING btree (passenger_id)
2	uq_passport_number	CREATE UNIQUE INDEX uq_passport_number ON public.passengers USING btree (passport_number)
3	unq_inf	CREATE UNIQUE INDEX unq_inf ON public.passengers USING btree (first_name, last_name, date_of_birth)
4	uq_passengers_passport	CREATE UNIQUE INDEX uq_passengers_passport ON public.passengers USING btree (passport_number)

Total rows: 4 Query complete 00:00:00.138

✓ Successfully run. Total query runtime: 138 msec. 4 rows affected. X

LF Ln 42, Col 1

Object Explorer

- Columns (14)
 - flight_id
 - flight_no
 - scheduled_departure
 - scheduled_arrival
 - departure_airport_id
 - arrival_airport_id
 - departing_gate
 - arriving_gate
 - airline_id
 - status
 - actual_departure
 - actual_arrival
 - created_at
 - update_at
- Constraints
- Indexes
- RLS Policies
- Rules
- Triggers
- passengers
 - Columns (10)
 - passenger_id
 - first_name
 - last_name
 - date_of_birth
 - gender
 - country_of_citizenship
 - country_of_residence
 - passport_number
 - created_at
 - update_at
 - Constraints
 - Indexes
 - RLS Policies
 - Rules

airport_DB/emilakshabayev@s

Query

```
40 from pg_indexes
41 where schemaname = 'public' and tablename = 'passengers';
42
43
44 insert into passengers (passenger_id, first_name, last_name, date_of_birth, gender,
45 country_of_citizenship, country_of_residence, passport_number,
46 created_at, update_at)
47 values (90001, 'test', 'ok', date '1990-01-01', 'Male',
48 'Kazakhstan', 'Kazakhstan', 'KZ-PASS-00090001', now(), now());
49
50 insert into passengers (passenger_id, first_name, last_name, date_of_birth, gender,
51 country_of_citizenship, country_of_residence, passport_number,
52 created_at, update_at)
53 values (90002, 'test', 'dup', date '1992-02-02', 'Female',
54 'Kazakhstan', 'Kazakhstan', 'KZ-PASS-00090001', now(), now());
55
```

Data Output Messages Notifications

INSERT 0 1

Query returned successfully in 40 msec.

Total rows: Query complete 00:00:00.040

LF Ln 48, Col 1

Object Explorer

- Columns (14)
 - flight_id
 - flight_no
 - scheduled_departure
 - scheduled_arrival
 - departure_airport_id
 - arrival_airport_id
 - departing_gate
 - arriving_gate
 - airline_id
 - status
 - actual_departure
 - actual_arrival
 - created_at
 - update_at
- Constraints
- Indexes
- RLS Policies
- Rules
- Triggers
- passengers
 - Columns (10)
 - passenger_id
 - first_name
 - last_name
 - date_of_birth
 - gender
 - country_of_citizenship
 - country_of_residence
 - passport_number
 - created_at
 - update_at
 - Constraints
 - Indexes
 - RLS Policies
 - Rules

airport_DB/emilakshabayev@s

Query

```
46 values (90001, 'test', 'ok', date '1990-01-01', 'Male',
47 'Kazakhstan', 'Kazakhstan', 'KZ-PASS-00090001', now(), now());
48
49 insert into passengers (passenger_id, first_name, last_name, date_of_birth, gender,
50 country_of_citizenship, country_of_residence, passport_number,
51 created_at, update_at)
52 values (90002, 'test', 'dup', date '1992-02-02', 'Female',
53 'Kazakhstan', 'Kazakhstan', 'KZ-PASS-00090001', now(), now());
54
55 --7
56 create index if not exists idx_passengers_cc_dob_lf
57 on passengers (country_of_citizenship, date_of_birth, last_name, first_name);
58
59 explain analyze
60 select passenger_id, first_name, last_name, date_of_birth,

```

Data Output Messages Notifications

INSERT 0 1

Query returned successfully in 58 msec.

Total rows: Query complete 00:00:00.058

LF Ln 53, Col 62

Object Explorer

- Columns (14)
 - flight_id
 - flight_no
 - scheduled_departure
 - scheduled_arrival
 - departure_airport_id
 - arrival_airport_id
 - departing_gate
 - arriving_gate
 - airline_id
 - status
 - actual_departure
 - actual_arrival
 - created_at
 - update_at
- Constraints
- Indexes
- RLS Policies
- Rules
- Triggers
- passengers
 - Columns (10)
 - passenger_id
 - first_name
 - last_name
 - date_of_birth
 - gender
 - country_of_citizenship
 - country_of_residence
 - passport_number
 - created_at
 - update_at
 - Constraints
 - Indexes
 - RLS Policies
 - Rules

airport_DB/emilakshabayev@s

Query

```
53 'Kazakhstan', 'Kazakhstan', 'KZ-PASS-0009001', now(), now();
54
55 --7
56 create index if not exists idx_passengers_cc_dob_lf
57 on passengers (country_of_citizenship, date_of_birth, last_name, first_name);
58
59 explain analyze
60 select passenger_id, first_name, last_name, date_of_birth
61 from passengers
62 where country_of_citizenship = 'Philippines'
63 and date_of_birth between date '1984-01-01' and date '1984-12-31'
64 limit 50;
65
66 --8
67 select indexname, indexdef
```

Data Output Messages Notifications

CREATE INDEX

Query returned successfully in 39 msec.

Total rows: Query complete 00:00:00.039

Query returned successfully in 39 msec. X

LF Ln 58, Col 1

Object Explorer

- Columns (14)
 - flight_id
 - flight_no
 - scheduled_departure
 - scheduled_arrival
 - departure_airport_id
 - arrival_airport_id
 - departing_gate
 - arriving_gate
 - airline_id
 - status
 - actual_departure
 - actual_arrival
 - created_at
 - update_at
- Constraints
- Indexes
- RLS Policies
- Rules
- Triggers
- passengers
 - Columns (10)
 - passenger_id
 - first_name
 - last_name
 - date_of_birth
 - gender
 - country_of_citizenship
 - country_of_residence
 - passport_number
 - created_at
 - update_at
 - Constraints
 - Indexes
 - RLS Policies
 - Rules

airport_DB/emilakshabayev@s

Query

```
55 --7
56 create index if not exists idx_passengers_cc_dob_lf
57 on passengers (country_of_citizenship, date_of_birth, last_name, first_name);
58
59 explain analyze
60 select passenger_id, first_name, last_name, date_of_birth
61 from passengers
62 where country_of_citizenship = 'Philippines'
63 and date_of_birth between date '1984-01-01' and date '1984-12-31'
64 limit 50;
65
66 --8
67 select indexname, indexdef
68 from pg_indexes
69 where schemaname = 'public' and tablename = 'passengers'
```

Data Output Messages Notifications

Showing rows: 1 to 6 Page No: 1 of 1

QUERY PLAN

1	Limit (cost=0.00..6.54 rows=1 width=23) (actual time=0.019..0.069 rows=1 loops=1)
2	→ Seq Scan on passengers (cost=0.00..6.54 rows=1 width=23) (actual time=0.017..0.067 rows=1 loops=1)
3	Filter: ((date_of_birth >= '1984-01-01':date) AND (date_of_birth <= '1984-12-31':date) AND ((country_of_citizenship)-text = 'Philippines'...
4	Rows Removed by Filter: 201
5	Planning Time: 1.210 ms
6	Execution Time: 0.095 ms

Total rows: 6 Query complete 00:00:00.040

LF Ln 65, Col 1

Object Explorer

- Columns (14)
 - flight_id
 - flight_no
 - scheduled_departure
 - scheduled_arrival
 - departure_airport_id
 - arrival_airport_id
 - departing_gate
 - arriving_gate
 - airline_id
 - status
 - actual_departure
 - actual_arrival
 - created_at
 - update_at
- Constraints
- Indexes
- RLS Policies
- Rules
- Triggers
- passengers
 - Columns (10)
 - passenger_id
 - first_name
 - last_name
 - date_of_birth
 - gender
 - country_of_citizenship
 - country_of_residence
 - passport_number
 - created_at
 - update_at
 - Constraints
 - Indexes
 - RLS Policies
 - Rules

Query

```

select passenger_id, first_name, last_name, date_of_birth
from passengers
where country_of_citizenship = 'Philippines'
and date_of_birth between date '1984-01-01' and date '1984-12-31'
limit 50;

--8
select indexname, indexdef
from pg_indexes
where schemaname = 'public' and tablename = 'passengers'
order by indexname;

drop index if exists uq_passengers_passport;
drop index if exists idx_passengers_cc_dob_lf;

```

Data Output

	indexname	indexdef
1	idx_passengers_cc_dob_lf	CREATE INDEX idx_passengers_cc_dob_lf ON public.passengers USING btree (country_of_citizenship, date_of_birth, last_name, first_name)
2	passengers_pkey	CREATE UNIQUE INDEX passengers_pkey ON public.passengers USING btree (passenger_id)
3	unq_inf	CREATE UNIQUE INDEX unq_inf ON public.passengers USING btree (first_name, last_name, date_of_birth)
4	uq_passengers_passport	CREATE UNIQUE INDEX uq_passengers_passport ON public.passengers USING btree (passport_number)
5	uq_passport_number	CREATE UNIQUE INDEX uq_passport_number ON public.passengers USING btree (passport_number)

Total rows: 5 Query complete 00:00:00.052

Object Explorer

- Columns (14)
 - flight_id
 - flight_no
 - scheduled_departure
 - scheduled_arrival
 - departure_airport_id
 - arrival_airport_id
 - departing_gate
 - arriving_gate
 - airline_id
 - status
 - actual_departure
 - actual_arrival
 - created_at
 - update_at
- Constraints
- Indexes
- RLS Policies
- Rules
- Triggers
- passengers
 - Columns (10)
 - passenger_id
 - first_name
 - last_name
 - date_of_birth
 - gender
 - country_of_citizenship
 - country_of_residence
 - passport_number
 - created_at
 - update_at
 - Constraints
 - Indexes
 - RLS Policies
 - Rules

Query

```

select passenger_id, first_name, last_name, date_of_birth
from passengers
where country_of_citizenship = 'Philippines'
and date_of_birth between date '1984-01-01' and date '1984-12-31'
limit 50;

--8
select indexname, indexdef
from pg_indexes
where schemaname = 'public' and tablename = 'passengers'
order by indexname;

drop index if exists uq_passengers_passport;
drop index if exists idx_passengers_cc_dob_lf;

```

Data Output

DROP INDEX

Query returned successfully in 43 msec.

Total rows: Query complete 00:00:00.043

Query returned successfully in 43 msec.